



FASR Science Case # WG1-1

Where is the Primary Electron Acceleration Site in Solar Flares?

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1 Science Goal(s)

Briefly summarize the key science goal(s) for this science case. A few sentences will be sufficient.

Solar flares are the most powerful explosions in the solar system and, at the same time, very efficient particle accelerators. Hence, they serve as an excellent “laboratory” to understand how charged particles are accelerated to nonthermal energies. This science case aims to use ultra-wideband radio imaging spectroscopy observations offered by FASR to address the question of where and how energetic electrons are accelerated in solar flares.

2 Scientific Rationale

2.1 Scientific Importance

Provide a brief discussion on the scientific importance of this science case.

Where and how energetic electrons are accelerated in solar flares remains one of the most debated topics in solar flare physics. Candidates of the electron acceleration site include the reconnection X line(s), the reconnection outflows, the above-the-looptop cusp region, the top of the flare arcade, and the flare arcade itself. Mechanisms involved in accelerating the electrons include stochastic acceleration by turbulence or cascading waves, merging magnetic islands, shocks, collapsing traps, large-scale Alfvén waves, and DC electric field. Despite significant progress over the past few decades, thanks to the advent of spaceborne and ground-based hard X-ray and radio instrumentation such as RHESSI and EOVS, we have been unable to pinpoint the primary acceleration site for flares with various sizes in a routine manner.

2.2 Uniqueness to FASR Capabilities

Is this science case uniquely addressed by FASR? Why can't other facilities address this science and achieve the same goal?

The decisive measurement is a comparison of the first nonthermal emission and its subsequent evolution at the candidate sites. A source that appears first at the reconnection region and then

spreads along the newly formed loops supports local acceleration there; a source that appears first in a looptop, outflow, or shock region supports a different hypothesis. The comparison requires the faint extended component to be recovered beside the bright looptop source, with enough frequency information to fit the electron spectrum at each location.

Energetic electrons are expected throughout the flare, but the regions outside the arcade are fainter and can be hidden by a compact looptop source. Modeling indicates that a dynamic range near $\sim 1000 : 1$ is needed to recover the energy-bearing component in the extended region (Chen et al. 2023). FASR is suited to this test because it combines broadband imaging spectroscopy with the field of view and angular resolution needed to compare the candidate sites in one event. The requirement is accepted only if source-injection tests recover the faint component and its fitted spectrum beside the bright source.

2.3 Synergies

Describe potential synergies/complementarities between this FASR science case and those from current/future/planned facilities at all wavelengths (e.g., DKIST, MUSE, FIERCE, COSMO, ngGONG, etc.).

FASR diagnostics of the flare-accelerated energetic electrons will critically complement those made by the current and potential hard X-ray instruments, including SolO/STIX and FIERCE, which primarily probe energetic electrons of < 100 keV in regions with a relatively high ambient density, while radio diagnostics also include > 100 keV electrons, weighted more strongly by the background magnetic field. In addition, the ability of FASR to outline magnetic loops/field lines “illuminated” by energetic electrons in a broad flaring region also allows detailed tracing of the electrons from the coronal region to the lower atmosphere, enabling detailed comparison with EUV and optical/IR observations made by, e.g., IRIS, MUSE (future), BBSO/GST, and DKIST.

2.4 Measurements Required by FASR

Provide a description of the necessary measurements to be carried out by the FASR to adequately address this science case. Please coordinate these measurements with the Science Requirements table in Section III.

Obtain co-temporal Stokes I and V imaging spectroscopy from 1–20 GHz, with approximately one-second cadence during the impulsive phase and no worse than 2

III. Science Requirements Tables

(A) Observational Target Description

Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.

(A) OBSERVATIONAL TARGET		
Type of observation (what defines a ‘target’)	<i>Provide a brief description of the target.</i> Spatially resolved gyrosynchrotron emission from the flare acceleration region, surrounding loops, and any connected current-sheet or looptop source.	
Number of targets	1 to few (number of flare sources across the disk)	
Size of a single target (arcsec x arcsec)	Up to $300'' \times 300''$	
Distribution of all targets (arcmin x arcmin)	Up to $30' \times 30'$	
Peak brightness (sfu/beam or Kelvin)	100–1000 MK in the brightest modeled looptop sources; event dependent	
RMS brightness (sfu/beam or Kelvin)	Science baseline: recover a 10^{-3} -of-peak extended component at $\geq 5\sigma$; the corresponding Kelvin value is set by the source-injection model	
Expected circularly polarized flux density	Stokes V (sfu/beam or Kelvin)	–
	V/I	Source and frequency dependent; the sign and spectral variation are required, while a fixed polarization fraction is not assumed
Expected linearly polarized flux density	Stokes Q or U (sfu/beam or Kelvin)	–
	Q/I or U/I	N/A for the baseline; linear polarization is not used to identify the acceleration site
hline		

(B) Spectro-Temporal Requirements Discussion

Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.

(B) SPECTRAL-TEMPORAL REQUIREMENTS	
Central Frequency (GHz)	1–20 GHz coverage; no single center frequency
hline Instantaneous Bandwidth (GHz/pol)	Sufficient tuning or processing to obtain the full 1–20 GHz science spectrum within each cadence
hline Spectral resolution [MHz]	$\Delta\nu/\nu \lesssim 2\%$ for the continuum; finer channels retained for coherent bursts
hline Temporal resolution (in seconds)	~ 1 s during the impulsive phase
hline	

(C) Polarization Product Discussion

Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.

(C) POLARIZATION DATA PRODUCTS REQUIRED	
y	Stokes I
n	Stokes Q
n	Stokes U
y	Stokes V

(D) Imaging Requirements Discussion

Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.

See discussions in Appendix.

(D) IMAGING REQUIREMENTS	
Required angular resolution (arcsec) (single value or range)	Resolve the compact looptop/current-sheet source from the surrounding flare region; target $\lesssim 2''$ near 10 GHz, with source-injection validation at lower frequencies
hline Largest angular scale required (arcsec)	300''
Mapped image size (arcmin x arcmin)	30' $\times$ 30'
Required pixel resolution (arcsec)	Sample the synthesized beam with several pixels at each frequency
hline Number of output/image channels	At least the number needed for $\Delta\nu/\nu \lesssim 2\%$ over 1–20 GHz; no fixed channel count is a baseline
hline Output bandwidth (minimum and maximum frequency - GHz)	1–20 GHz
Channel width (MHz)	Set by the fractional-spacing requirement; finer low-frequency channels are retained for coherent bursts
hline Required rms (sfu/beam or Kelvin) [per channel] (if polarization products required define for each)	Recover a 10^{-3} -of-peak extended component at $\geq 5\sigma$ in Stokes I and V; numerical temperature follows the source model
hline Dynamic range within image (if polarization products required define for each)	$\geq 1000 : 1$, validated by recovery of the faint extended component beside the brightest flare source

(D) IMAGING REQUIREMENTS (continued)		
hline Polarization accuracy (%)	Enough absolute Stokes-V accuracy to recover the sign and spectral evolution of the injected component; 2 percentage points is a goal for bright sources	
hline Zero spacing/total power required?	Preferred for absolute energetics and extended emission; compact-source localization can proceed from interferometric imaging	
hline Required maximum latency (in seconds, or N/A)	N/A	
Required flux density scale calibration accuracy		1-3%
	x	5%
		10%
		20-50%

2.5 Other Performance or Functional Requirements

If there are any additional performance or functional requirements not captured above, describe them here. For example, beamforming array mode, phased array, etc.

3 Appendix

Please provide any other relevant material necessary to understand and substantiate this Science Case.

The dynamic-range requirement comes from flare models that place synthetic microwave sources at the looptop, current sheet, and flux-rope regions. The acceptance test should inject those sources into representative FASR visibilities and require recovery of the faint source location, spectral slope, and onset time while the bright source is present. This test defines the needed thermal-noise and calibration margin; a standalone Kelvin target should not be adopted before the source model and image weighting are fixed.

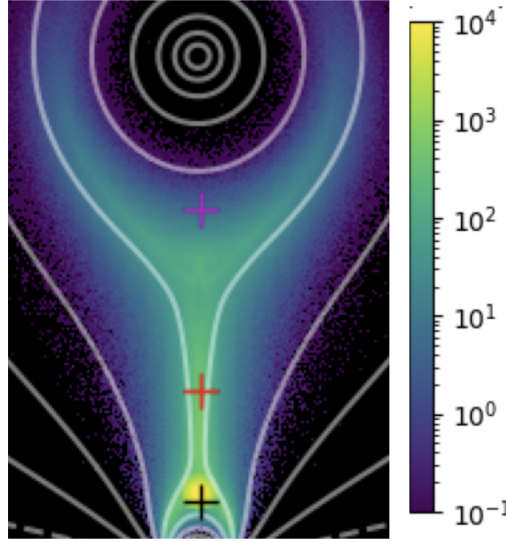
The continuum sampling is specified as a fractional frequency spacing because the fitted spectral curvature, rather than one fixed channel width, constrains the electron distribution. A 2% spacing gives comparable fractional leverage across 1–20 GHz. The low-frequency burst product is separate: its channel width must be set by the observed fine structure and verified with source-injection tests.

Stokes V is required because its sign and frequency dependence provide an independent constraint on the emitting electrons and magnetic-field orientation. The exact fractional polarization varies with optical depth, viewing angle, and source mixture, so the baseline is stated as recovery of the injected V spectrum rather than as a universal V/I value. A 2-percentage-point error is a goal for

bright sources.

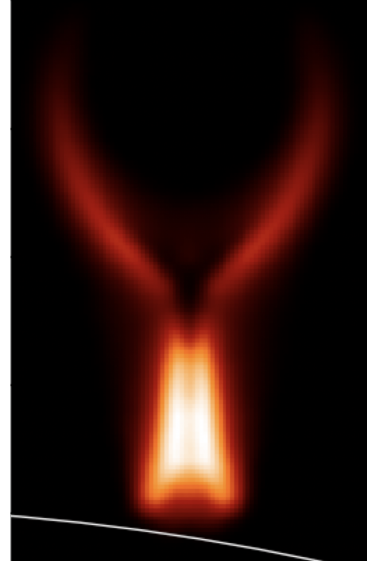
The requirement of time resolution of ~ 1 s is chosen to capture “snapshots” of the distribution of energetic electrons. Recent modeling (e.g., Xiaocan Li et al. 2022) results suggest that the overall, macroscopic distribution of flare-accelerated electron distribution evolves at MHD scales. With an angular resolution of $2''$ at 10 GHz (mid-frequency) and a characteristic Alfvén speed of 1000 km/s, this gives us a typical timescale of 1.5 s. Therefore, a time cadence of 1 s is sufficient to study the collective evolution in most cases, although a much higher time resolution is desired to decouple the transport effects from acceleration, which is described in another science case.

>100 keV Electron Distribution

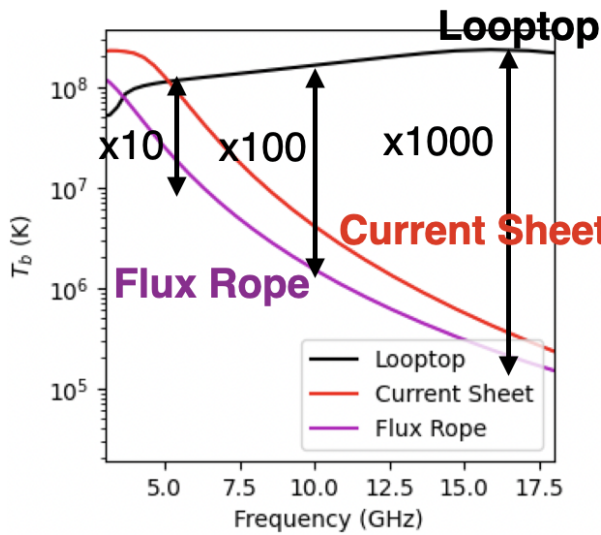


Simulation by Xiaocan Li, Fan Guo et al

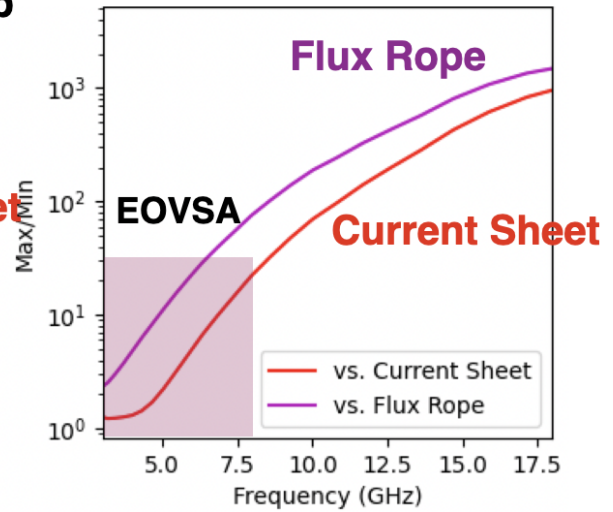
Simulated FASR



Microwave Spectra



Dynamic Range Requirement



Reference

- Chen, B., Gary, D., Yu, S., Mondal, S., Fleishman, G., Li, X., Shen, C., Guo, F., et al. 2023. “Quantifying Energy Release in Solar Flares and Solar Eruptive Events: New Frontiers with a Next-Generation Solar Radio Facility.” *Bulletin of the American Astronomical Society*, 55, 060. <https://doi.org/10.3847/25c2cfef.aa2ad1d0>.
- Li, X., Guo, F., Chen, B., Shen, C., and Glesener, L. 2022. “Modeling Electron Acceleration and Transport in the Early Impulsive Phase of the 2017 September 10th Solar Flare.” *The Astrophysical Journal*, 932, 92. <https://doi.org/10.3847/1538-4357/ac6efe>.